

Microsoft Excel

A Practical Guide to Cells, Formulas, and Data Tools

Microsoft Excel is the standard tool for working with numbers, data, and calculations in a spreadsheet format. This guide covers the basics of cells and worksheets, how to build formulas and use functions, formatting data, tables, charts, and other tools that make Excel powerful for everything from a simple budget to complex data analysis.

1. Understanding Cells, Rows, and Columns

An Excel worksheet is a grid made of **cells**, each identified by its column letter and row number (e.g., **A1**, **B2**). A group of cells is called a **range**, written as the first and last cell separated by a colon (e.g., **A1:A10**).

1.1 Selecting and Entering Data

- 1 Click a cell to select it, or click-and-drag to select a range of cells.
- 2 Type a value and press Enter (moves down) or Tab (moves right) to confirm.
- 3 Double-click a cell, or press F2, to edit its existing contents.

1.2 Rows & Columns

- 1 Right-click a row/column header and choose **Insert** or **Delete** to add or remove one.
 - 2 Drag the border of a column header to resize its width; double-click the border to auto-fit.
 - 3 Right-click a row/column header and choose **Hide** to temporarily hide it without deleting data.
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2. Writing Formulas

Every formula in Excel begins with an equals sign (=). Once entered, Excel calculates the result automatically and updates it whenever the referenced cells change.

2.1 Basic Arithmetic

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=A1+B1 (addition)
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=A1-B1 (subtraction)
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=A1*B1 (multiplication)

=A1/B1 (division)

=A1^2 (exponent / power)

2.2 Order of Operations

Excel follows standard math order of operations (parentheses, exponents, multiplication/division, then addition/subtraction). Use parentheses to control calculation order, e.g. **=(A1+B1)*C1**.

2.3 Cell References: Relative vs. Absolute

By default, references are **relative** — they shift automatically when a formula is copied to another cell. Adding a dollar sign makes a reference **absolute**, so it stays fixed when copied.

- **A1** — relative reference (changes when copied)
- **\$A\$1** — absolute reference (stays fixed when copied)
- **\$A1** or **A\$1** — mixed reference (locks only the column or only the row)

Tip: Press F4 after clicking a cell reference inside a formula to quickly cycle through relative, absolute, and mixed reference styles.

3. Common Functions

Functions are pre-built formulas for common calculations. All start with an equals sign followed by the function name and its inputs ("arguments") in parentheses.

Function	Purpose	Example
SUM	Adds a range of numbers	=SUM(A1:A10)
AVERAGE	Calculates the mean	=AVERAGE(A1:A10)
MIN / MAX	Finds the smallest/largest value	=MIN(A1:A10)
COUNT / COUNTA	Counts numeric / non-empty cells	=COUNTA(A1:A10)
IF	Returns a value based on a condition	=IF(A1>50, "Pass", "Fail")
VLOOKUP	Looks up a value in the first column of a range	=VLOOKUP(A1, D1:F10, 2, FALSE)
XLOOKUP	Modern, flexible lookup (newer Excel versions)	=XLOOKUP(A1, D:D, E:E)
SUMIF / SUMIFS	Sums values that meet one/multiple conditions	=SUMIF(A1:A10, ">50")
COUNTIF / COUNTIFS	Counts cells that meet one/multiple conditions	=COUNTIF(A1:A10, "Yes")
CONCAT / TEXTJOIN	Combines text from multiple cells	=TEXTJOIN(" ", TRUE, A1, B1)
ROUND	Rounds a number to a set number of digits	=ROUND(A1, 2)
TODAY / NOW	Returns the current date / date and time	=TODAY()

Tip: Click the fx button next to the formula bar to browse all available functions with a description and guided input for each argument.

4. Copying Formulas & AutoFill

- 1 Select a cell containing a formula.
 - 2 Hover over the small square at the bottom-right corner of the cell (the **fill handle**) until the cursor becomes a thin cross.
 - 3 Click and drag down or across to copy the formula to adjacent cells — relative references adjust automatically.
 - 4 Double-click the fill handle to automatically fill down to match the length of an adjacent column of data.
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5. Formatting Cells

5.1 Number Formats

- 1 Select the cell(s) you want to format.
- 2 On the Home tab, use the Number Format dropdown to choose Currency, Percentage, Date, etc.
- 3 For more control, press Ctrl+1 to open the full Format Cells dialog (decimal places, currency symbol, date style, etc.).

5.2 Fonts, Borders, and Fill Color

- 1 Use the Home tab's Font group to change font type, size, bold/italic, and text color.
- 2 Use the Borders dropdown to add gridlines or box borders around selected cells.
- 3 Use the Fill Color bucket icon to shade cells (useful for highlighting headers or key figures).

5.3 Conditional Formatting

Automatically applies formatting (colors, icons, data bars) to cells based on their value, making it easy to spot trends or outliers at a glance.

- 1 Select the range you want to apply formatting to.
- 2 Go to Home > Conditional Formatting.
- 3 Choose a rule type, such as "Highlight Cell Rules," "Color Scales," or "Data Bars."
- 4 Set the condition (e.g., "Greater Than 100") and choose a formatting style.

6. Tables, Sorting & Filtering

6.1 Converting a Range to a Table

- 1 Select your data range (including headers).
- 2 Go to Insert > Table (or press Ctrl+T).
- 3 Confirm the range and check "My table has headers," then click OK.
- 4 Excel applies banded formatting and adds dropdown filter arrows to each header automatically.

6.2 Sorting Data

- 1 Click any cell within the column you want to sort by.
- 2 Go to Data > Sort A-Z (ascending) or Z-A (descending), or use Data > Sort for multi-level sorting.

6.3 Filtering Data

- 1 Click the dropdown arrow in a column header (added automatically for tables, or via Data > Filter).
 - 2 Uncheck values you want to hide, or use Text/Number Filters for custom conditions.
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7. Creating Charts

- 1 Select the data range you want to chart, including headers.
- 2 Go to the **Insert** tab and choose a chart type from the Charts group (Column, Line, Pie, etc.).
- 3 Use the Chart Design tab to change the chart style, colors, or switch chart type.
- 4 Click the + icon next to the chart to add or remove elements like axis titles, data labels, or a legend.

Tip: Select your data and press Alt+F1 for an instant default chart, which you can then customize.

8. Summarizing Data with PivotTables

A PivotTable lets you quickly summarize, group, and analyze large datasets without writing formulas.

- 1 Click anywhere inside your data range.
 - 2 Go to Insert > PivotTable, and choose where to place it (new worksheet is recommended).
 - 3 In the PivotTable Fields pane, drag fields into the **Rows**, **Columns**, **Values**, and **Filters** boxes.
 - 4 By default, numeric fields dropped into Values are summed; click the field to change it to Average, Count, etc.
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9. Other Useful Tools

9.1 Freeze Panes

Keeps header rows/columns visible while scrolling through large datasets. Go to View > Freeze Panes > Freeze Top Row (or Freeze First Column).

9.2 Data Validation

Restricts what can be entered into a cell, useful for building forms or preventing input errors. Go to Data > Data Validation, then set allowed criteria (e.g., a dropdown list, whole numbers only, or a date range).

9.3 Find & Replace

Press Ctrl+F to find data or Ctrl+H to find and replace values across the worksheet.

9.4 Named Ranges

Select a range, then type a name into the Name Box (top-left, next to the formula bar) to reference that range by name in formulas instead of cell coordinates, e.g. **=SUM(Sales)**.

10. Common Formula Errors

Error	Meaning	Typical Fix
#DIV/0!	Formula is dividing by zero	Check the divisor cell isn't empty or zero
#N/A	A lookup value wasn't found	Verify the lookup value exists and matches exactly
#REF!	A cell reference is invalid	Occurs after deleting a referenced cell/row/column
#NAME?	Excel doesn't recognize text in the formula	Check for typos in a function name
#VALUE!	Wrong type of argument or operand	Ensure text isn't being used where a number is expected

11. Summary

Excel's power comes from combining simple building blocks: cells organized into rows and columns, formulas that calculate automatically, and functions that handle common tasks like summing, averaging, and looking up data. Layer on formatting, tables, charts, and PivotTables, and a plain grid of numbers becomes a fully interactive tool for tracking budgets, analyzing datasets, or building reports — all updating instantly whenever the underlying data changes.